

# Filtering germs

Germs that break through the body's barriers and invade internal tissues do not usually survive for long. The human body contains a network of tiny vessels that collect fluid from every type of tissue and carefully filter it for germs, which are then swiftly destroyed.

## LYMPHATIC SYSTEM

Fluid from body tissues is collected by the vessels of the lymphatic system. As it flows along lymphatic vessels, the fluid passes through swellings called lymph nodes. These are packed with white blood cells called lymphocytes, which identify germs and kill them.

*After being filtered, fluid from the lymphatic system enters the bloodstream here.*

*Fluid enters here.*

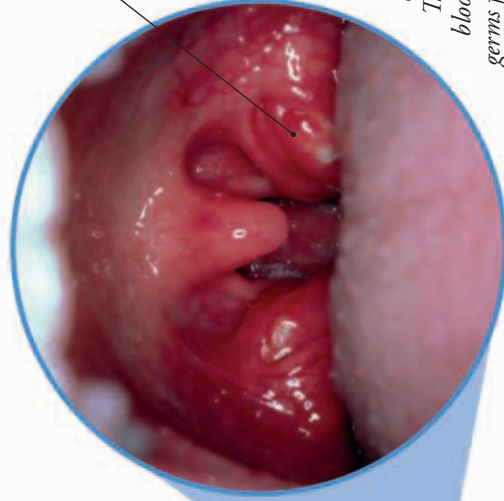
### ► LYMPH NODE

*Fluid is filtered as it passes through lymph nodes, which vary in size from a period to a grape.*

*New lymphocytes are produced here.*

*Lymphocyte*

*Cleaned fluid leaves here.*



*Tonsils swell up when germs have caused an infection nearby.*

### ▼ TONSILS

*At the back of the mouth are small organs called tonsils. These contain white blood cells that destroy germs from food or the air.*

*The thymus gland in the chest produces white blood cells and helps train them to attack germs but not healthy body cells.*

## FAST FACTS

- The lymphatic system produces germ-killing chemicals called antibodies.
- The white blood cells in lymph nodes are able to remember germs and mount attacks much more swiftly if they encounter them in the future.
- Lymph organs such as the tonsils and thymus are large during childhood and wither away as we get older.
- The spleen is the largest lymph organ in the body.

*The spleen filters blood for germs. It also acts as a store for lymphocytes.*

